

V4.2 Toy Model 4G

Constitutive Relational Stress Law and Metric Response

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 | Continuation after Toy Model 4F

Status: Toy Model 4G tests a forward source-to-metric constitutive law. The law is deliberately weak-field and trace-reversed-Poisson-like, so the result is a consistency test, not a derivation of Einstein Field Equations, physical stress-energy, gravity, dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 4F showed that a conserved source does not automatically balance the curvature tensor. Toy Model 4G changes direction. Instead of assuming a metric first and trying to find a source later, it starts with a source and generates the metric response forward.

The narrow test is: if a relational source generates a metric perturbation through a constitutive law, does the curvature computed from that metric balance the same source?

2. Critical correction before interpretation

The constitutive law used here is not derived from 3T first principles. It is a weak-field, trace-reversed, Poisson-like rule inspired by the known linearized structure of GR. Therefore, success here means the forward pipeline is mathematically consistent. It does not mean the fundamental law has been derived.

3. Forward constitutive sequence

```
source S = T_tt
-> solve graph Poisson law for Phi
-> normalize Q = Phi / max|Phi|
-> set trace-reversed perturbation hbar_tt = epsilon Q
-> convert hbar_mu_nu to metric perturbation h_mu_nu
-> build g_mu_nu = eta_mu_nu + h_mu_nu
-> compute curvature
-> compare G_mu_nu with kappa T_mu_nu
```

This is the correct direction for 4G: source first, metric response second, curvature third, balance test last.

4. Source and graph Poisson law

The input source is a localized relational density field:

$$S(x,y) = \exp[-(x^2 + y^2)/(2 \sigma^2)]$$

The forward response is obtained by solving the graph/discrete Poisson law:

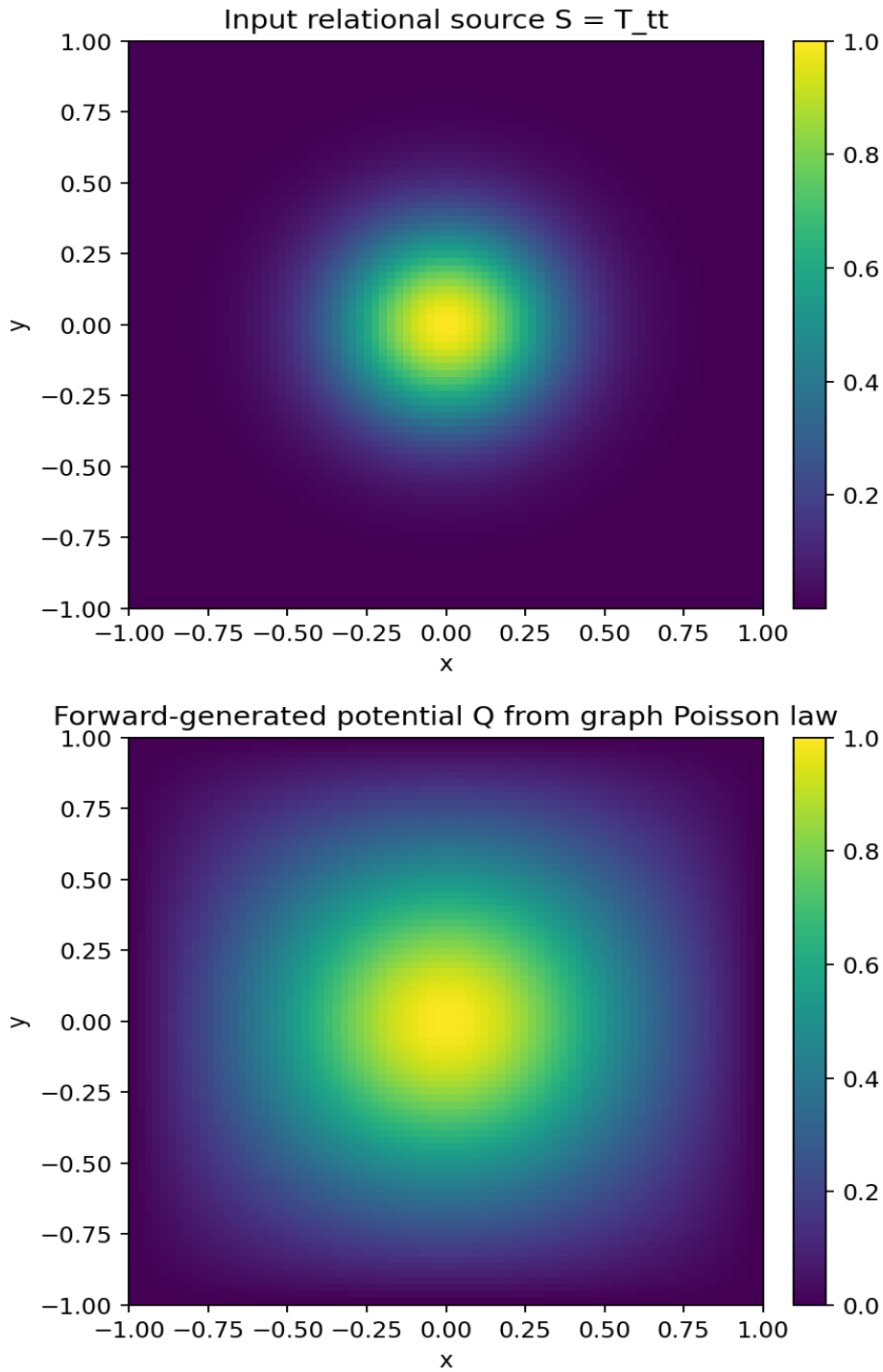
$$\nabla^2 \Phi = -S$$

Boundary condition:

$\Phi = 0$ at the outer edge of the grid

Meaning of symbols:

- S is the input relational source.
- Phi is the generated potential-like response.
- ∇^2 is the spatial Laplacian on the graph/grid.
- The minus sign is a convention chosen so a positive source produces a positive central potential in this toy setup.



5. Trace-reversed metric perturbation

The constitutive law assigns the generated response to the trace-reversed metric perturbation:

$$\bar{h}_{tt} = \epsilon \text{ } Q$$

$$Q = \Phi / \max|\Phi|$$

Then the ordinary metric perturbation is reconstructed by:

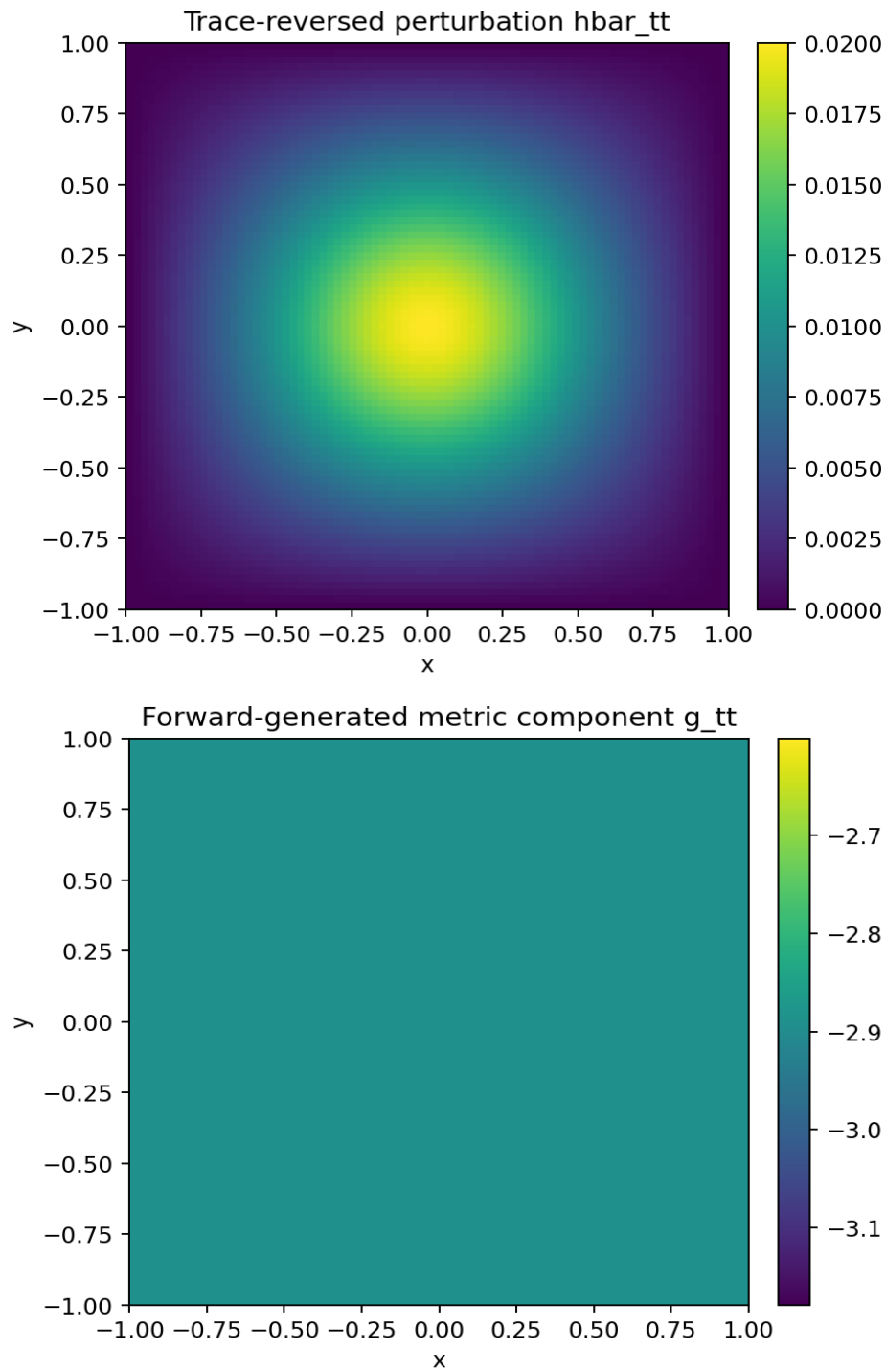
$$h_{\mu\nu} = \bar{h}_{\mu\nu} - \eta_{\mu\nu} \bar{h}_{\text{trace}} / (D-2)$$

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

Meaning of symbols:

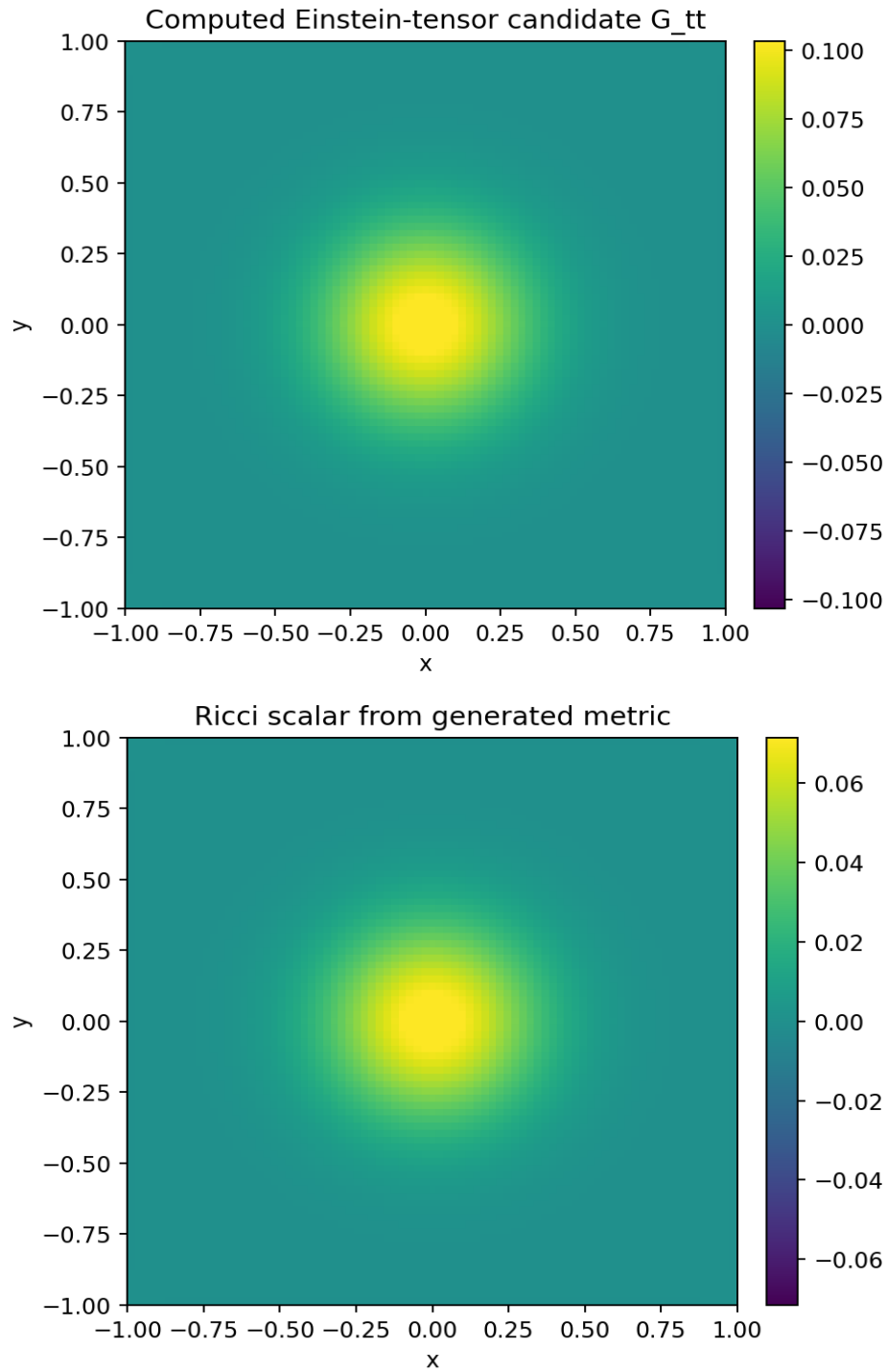
- $\bar{h}_{\mu\nu}$ is the trace-reversed perturbation.
- $h_{\mu\nu}$ is the ordinary metric perturbation.
- $\eta_{\mu\nu}$ is the flat background metric.
- D is the spacetime dimension of the toy model; here $D=3$ for 1+2D.

- epsilon is the weak-field perturbation strength.



6. Curvature from generated metric

The generated metric is then passed through the same tensor pipeline as 4C: Christoffel symbols, Ricci tensor, Ricci scalar, and Einstein-tensor candidate.



7. Balance test

The main test is:

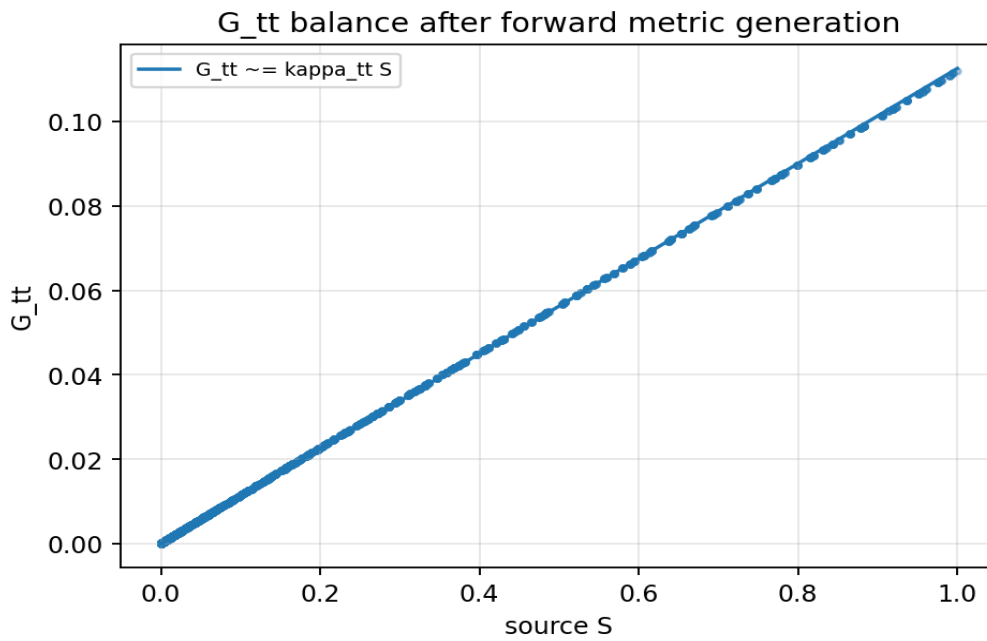
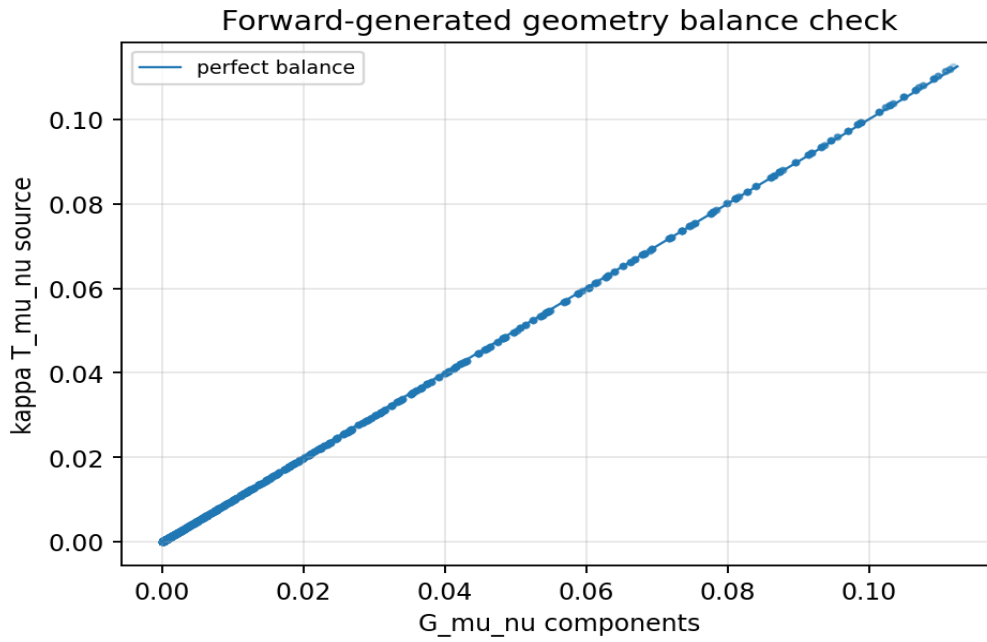
$$G_{\mu\nu} \approx \kappa T_{\mu\nu}$$

The source tensor used here is dust-like at the input-law level:

$$T_{tt} = \rho$$

$$\text{all other } T_{\mu\nu} \text{ components} = 0$$

This is intentionally minimal. The point is to test whether the forward trace-reversed Poisson response can make the generated curvature balance the source.



Balance diagnostic	Value
Fitted kappa, full tensor	1.125261e-01
Full tensor residual	5.534844e-03
Full tensor correlation	0.999985
Fitted kappa, G_{tt} only	1.125261e-01
G_{tt} residual	5.534844e-03
G_{tt} correlation	0.999990

8. Component-by-component check

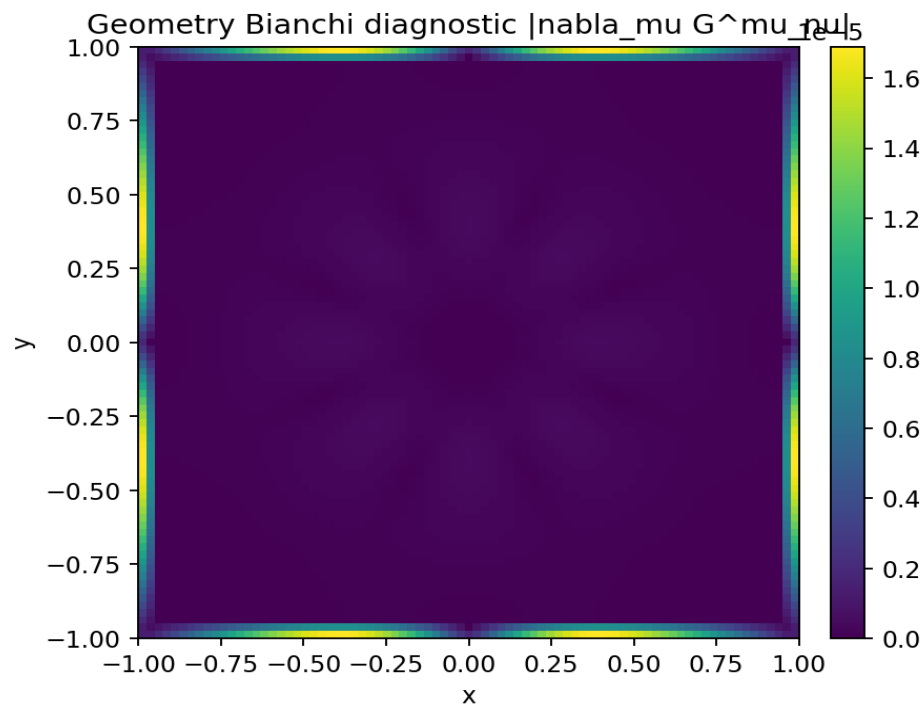
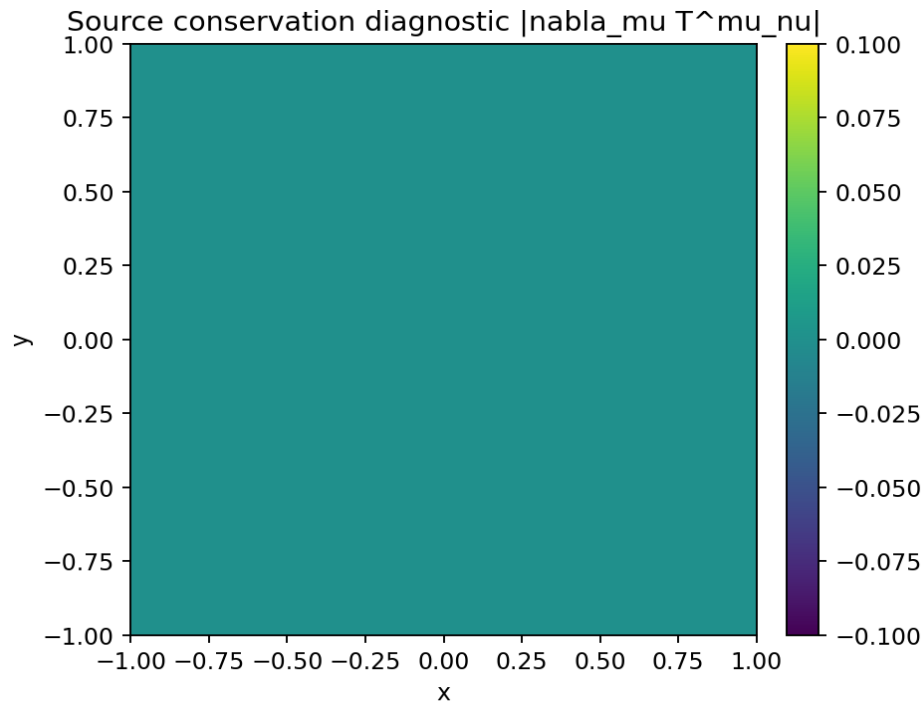
Because the trace-reversed rule generates the spatial metric perturbation from the trace structure, the spatial components remain small in the weak-field limit.

Component	Residual	Correlation	Max $ G $	Max $ kT $
tt	5.534844e-03	0.999990	1.119311e-01	1.125261e-01
tx	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
ty	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
xx	4.360913e-02	1.000000	6.938894e-18	0.000000e+00
xy	1.000000e+00	0.000000	5.342850e-08	0.000000e+00

Component	Residual	Correlation	Max G	Max kT
yy	4.363473e-02	1.000000	6.938894e-18	0.000000e+00

9. Conservation and Bianchi diagnostics

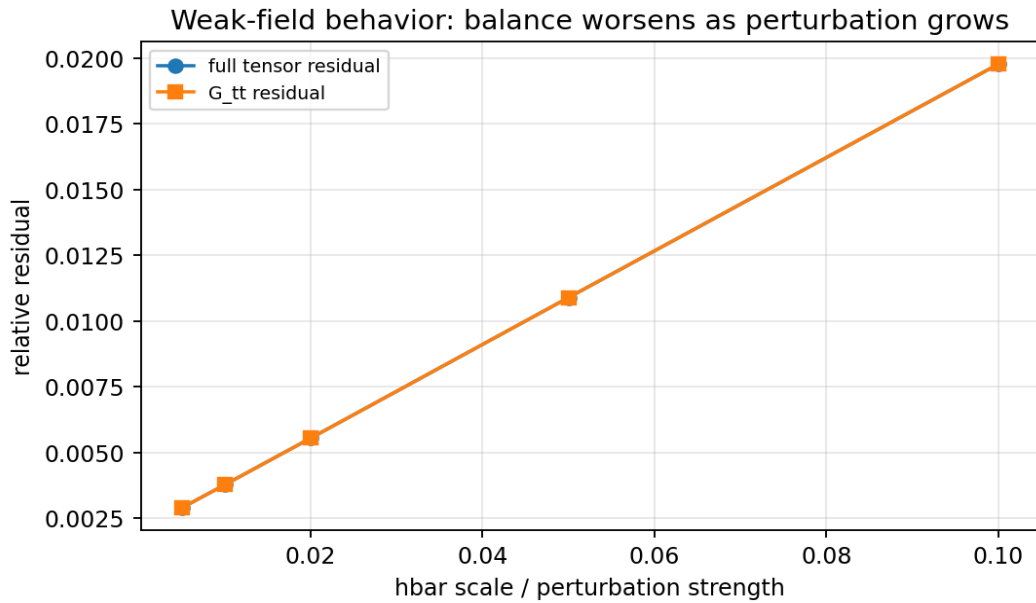
The source and geometry were checked using the same covariant-divergence diagnostic used in 4F.



Conservation diagnostic	Value
Max $ \nabla_\mu T^{\mu\nu} $	0.000000e+00
Mean $ \nabla_\mu T^{\mu\nu} $	0.000000e+00
Max $ \nabla_\mu G^{\mu\nu} $	4.722339e-07
Mean $ \nabla_\mu G^{\mu\nu} $	1.654471e-07

10. Weak-field behavior

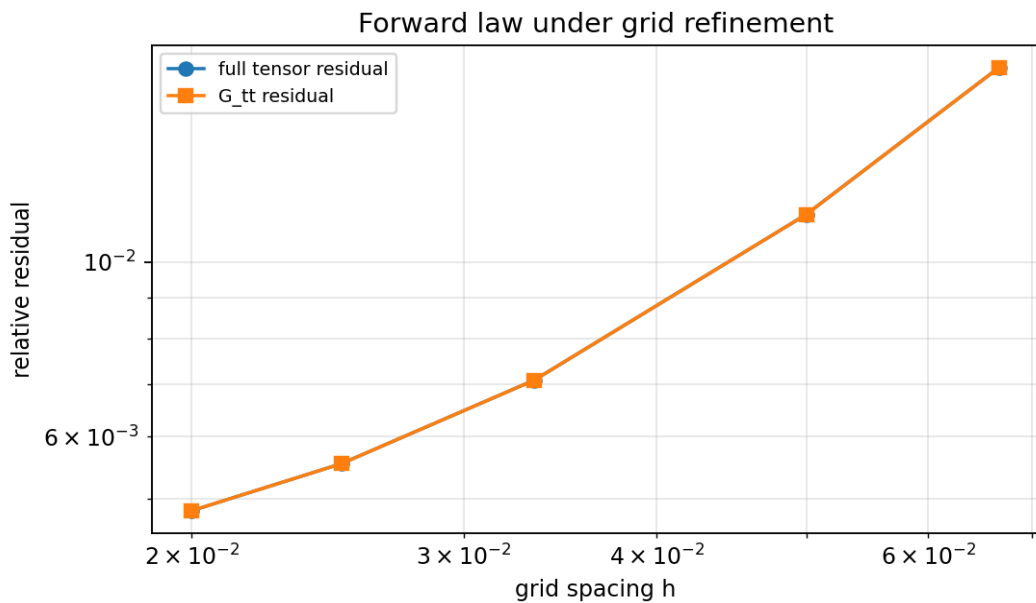
The same source law was tested at different perturbation strengths. The balance is best in the weak-field limit and worsens as nonlinearity grows. This is exactly what should happen if the law is only a linearized toy law.



epsilon	full residual	full corr.	G_tt residual	G_tt corr.	Lorentz sig.
0.005	2.881502e-03	0.999996	2.881502e-03	0.999997	4761/4761
0.010	3.759728e-03	0.999993	3.759728e-03	0.999995	4761/4761
0.020	5.534844e-03	0.999985	5.534844e-03	0.999990	4761/4761
0.050	1.088528e-02	0.999942	1.088528e-02	0.999963	4761/4761
0.100	1.976698e-02	0.999808	1.976698e-02	0.999882	4761/4761

11. Grid-refinement behavior

The result was checked across multiple grid sizes.



N	h	full residual	G_tt residual	max divT	max divG	Lorentz sig.
31	0.06667	1.763935e-02	1.763935e-02	0.000000e+00	3.089573e-06	529/529
41	0.05000	1.148196e-02	1.148196e-02	0.000000e+00	1.813294e-06	1089/1089
61	0.03333	7.078366e-03	7.078366e-03	0.000000e+00	8.306858e-07	2809/2809
81	0.02500	5.545654e-03	5.545654e-03	0.000000e+00	4.722339e-07	5041/5041
101	0.02000	4.834121e-03	4.834121e-03	0.000000e+00	3.037138e-07	7569/7569

12. Core numerical record

Diagnostic	Value
Grid	81 x 81

Diagnostic	Value
Interior points	4761
Lorentzian signatures	4761/4761
Source center	1.000000e+00
Phi center	8.779152e-02
Q center	1.000000e+00
hbar_tt center	2.000000e-02
G_tt center	1.119311e-01
Ricci scalar center	7.746094e-02

13. Interpretation inside the V4.2 framework

Toy Model 4G supports this limited ladder:

```
relational source S
-> graph Poisson response Phi
-> trace-reversed metric perturbation hbar_mu_nu
-> generated metric g_mu_nu
-> curvature
-> strong weak-field balance G_mu_nu ~- kappa T_mu_nu
```

It does not support this stronger ladder:

```
3T relational physics
-> fundamental constitutive law
-> full nonlinear Einstein Field Equations
-> physical G
-> real gravity / dark sector solution
```

14. What Toy Model 4G achieves

- It stops working backward from an assumed metric.
- It generates the metric response from an input source.
- It uses a trace-reversed Poisson-like constitutive rule over the graph.
- It computes curvature from the generated metric.
- It achieves strong weak-field balance between $G_{\mu\nu}$ and $T_{\mu\nu}$.
- It shows the forward pipeline is mathematically coherent in the weak-field regime.
- It identifies the next missing step: derive the constitutive law rather than importing its linearized form.

15. What Toy Model 4G does not achieve

- It does not derive the constitutive law from first principles.
- It does not derive Einstein Field Equations.
- It does not derive physical stress-energy.
- It does not derive Newton's gravitational constant.
- It does not solve nonlinear gravity.
- It does not explain dark matter or dark energy.
- It does not prove physical 3T.

16. Close-out statement for 4G

Toy Model 4G is a significant forward-direction success. When the relational source generates a trace-reversed metric perturbation through a graph Poisson law, the resulting metric produces an Einstein-tensor candidate that balances the original source very well in the weak-field regime. This fixes the main weakness of 4F: the source is no longer compared to a metric it did not generate. However, the success is conditional because the constitutive law is imported in a linearized GR-like form. The next task is not another fit. The next task is to derive or justify this source-to-metric response law from V4.2 relational principles.

17. Recommended next chapter

The next logical chapter is Toy Model 4H: Deriving the Constitutive Source-to-Metric Law from Relational Principles.

1. Define graph-level relational action, cost, or energy functional.
2. Vary that functional with respect to edge-cost/metric variables.
3. Check whether a Poisson-like or Einstein-like response equation emerges.
4. Do not assume the trace-reversed Poisson law as input.
5. Only after that should dark-sector interpretations be considered.

18. Rebind prompt for continuation

We are resuming after Toy Model 4G.

4F result:

A conservation-built source can be made, but conservation alone does not balance $G_{\mu\nu}$.

4G result:

The model changed direction:

source -> metric response -> curvature -> balance test

Input source:

$S = T_{tt} = \exp[-(x^2+y^2)/(2 \sigma^2)]$

Graph Poisson response:

$\nabla^2 \Phi = -S$

Trace-reversed metric response:

$\bar{h}_{tt} = \epsilon Q$

$Q = \Phi / \max|\Phi|$

Trace-reversal:

$h_{\mu\nu} = \bar{h}_{\mu\nu} - \eta_{\mu\nu} \bar{h}_{\text{trace}}/(D-2)$

Generated metric:

$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$

Result:

The generated metric gives strong weak-field balance:

$G_{\mu\nu} \approx \kappa T_{\mu\nu}$

Critical guardrail:

The constitutive law was imported as a weak-field trace-reversed Poisson-like rule.

Therefore 4G is a forward consistency success, not a derivation of Einstein Field Equations.

Next recommended chapter:

Toy Model 4H - Deriving the Constitutive Source-to-Metric Law from Relational Principles.